

Jindou Jia

Postdoctoral Researcher in Robot Learning, Nanyang Technological University

jindou.jia@ntu.edu.sg · jiajindou.com · JIAjindou · Google Scholar

Robot-learning researcher trained in control theory. I combine world models, generative policies and uncertainty prediction with physical priors, so that robots transfer reliably from training to environments they have never seen, and I validate this work on real hardware.

23 Papers · **18** First / co-first author · **5** Corresponding author · **13** Robot platforms

T-RO · IJRR · NeurIPS · ICLR Oral · RSS · CoRL · ICRA · npj Robotics · R-AL

Control and learning, in both directions

PhD in control science at Beihang University, now at the frontier of robot learning at NTU. **Learning for control:** EVOLVER (T-RO), FORESEER (IJRR). **Control for learning:** feedback neural ODEs (ICLR Oral), PhyFilter (npj Robotics), ODE policy.

Efficient policies and world models

Generative policies built for real-time control: single-step inference (FLASH at NeurIPS, Δ-Flow), action-to-action transport (A2A at RSS) and multimodality only when it matters (MARS). Feedback world models that correct predictions online (NeurIPS) and a world-model survey (IJRR).

Proven on real robots, across morphologies

Worked with 13 robot platforms, from agile quadrotors and grasping drones to robot arms, dual-arm systems, a quadruped and a humanoid. ODE policy was tested on 14 real-world tasks across single-arm, dual-arm, dexterous and humanoid robots.

Research leadership and open source

Project lead of ODE policy; corresponding author on 5 recent papers; Associate Editor of IEEE RA-L; ICCA 2024 Best Student Paper. Guest lecture at NUS, invited talks at NJU and SIMTech & ARTC. Code released for ODE policy, FLASH, PhyFilter, EVOLVER and more.

Robot learning	imitation learning, reinforcement learning, flow matching and diffusion policies, world models, vision-language-action models, sim-to-real
Control and planning	disturbance observers, agile flight control, uncertainty prediction, trajectory optimization, feedback for learned dynamics
Robots	quadrotors, grasping drones, robot arms, dual-arm systems, dexterous hands, quadruped, humanoid

Research Trajectory

01 **Anti-disturbance control** 2019 – 2022
02 **Learning disturbances** 2023 – 2026
03 **Uncertainty prediction & planning** 2024 – 2026
04 **Feedback & world models** 2025 –
05 **Efficient generative policies** 2025 –
06 **Physical priors for learning** 2026 –
→ **Generalizable embodied intelligence** Next



Experience

Nanyang Technological University Nov 2025 - now
Postdoc in the School of Mechanical and Aerospace Engineering, working with Prof. Jianfei Yang at the MARS Lab.

Hangzhou Innovation Institute, Beihang University Jun 2025 - Nov 2025
Research Fellow.

Shenzhen Taiyi Technology (SKYMAGIC) Aug 2021 & Aug 2022
Drone control intern.

Education

Beihang University Fall 2020 - Spring 2025
Ph.D. in Control Science and Control Engineering, advised by Prof. Lei Guo, Prof. Xiang Yu, and Prof. Kexin Guo.
Shenyuan Honors College (⩽3%).

Beihang University Fall 2019 - Spring 2020
M.S. in Control Science and Control Engineering.

Harbin Engineering University Fall 2015 - Spring 2019
B.S. in Automation.

Publications

[*] Equal contribution [†] Corresponding author

Selected Publications

Physics filtering favors the generalization of robot learning — npj Robotics'26
Jindou Jia[*], Shixuan Han[*], Meng Wang[*], G. Li, Z. Yang, S. Zhou, K. Guo[†], J. Yang[†], X. Yu, W. Wang, L. Guo

Action-to-action flow matching — RSS'26
Jindou Jia[*], Gen Li[*], X. Chen, T. An, Y. Hu, J. Li, X. Guo, J. Yang

FORESEER: Recognize and utilize uncertainties by integrating data-based learning and symbolic feedback — IJRR'25
Jindou Jia, K. Guo, Y. Wang, S. Zhou, J. Zhang, Y. Liu, X. Yu, Y. Shi, L. Guo

Feedback favors the generalization of neural ODEs — ICLR'25 Oral Presentation
Jindou Jia[*], Zihan Yang[*], M. Wang, K. Guo, J. Yang, X. Yu, L. Guo

EVOLVER: Online learning and prediction of disturbances for robot control — T-RO'24
Jindou Jia, W. Zhang, K. Guo, J. Wang, X. Yu, Y. Shi, L. Guo

Accurate high-maneuvering trajectory tracking for quadrotors: A drag utilization method — R-AL'22
Jindou Jia, K. Guo, X. Yu, W. Zhao, L. Guo

Other Peer-Reviewed Publications

ADAPT: Analytical disturbance-aware policy training for humanoid locomotion — CoRL'26
Bofan Lyu[*], **Jindou Jia[*],** K. Zuo, Y. Lu, S. Han, G. Li, B. Ma, J. Li, G. Li, J. Yang

Feedback world model enables precise guidance of diffusion policy — NeurIPS'26
Tuo An[*], **Jindou Jia[*],** G. Li, J. Li, C. Zhou, P. Liu, B. Lyu, J. Bai, X. Guo, G. Li, J. Yang

FLASH: Efficient visuomotor policy via sparse sampling — NeurIPS'26
Jiaqi Bai[*], **Jindou Jia[*],** Y. Hu, G. Li, X. Chen, T. An, K. Zuo, J. Yang

World model for robot learning: A comprehensive survey — IJRR'26
Bohan Hou[*], Gen Li[*], **Jindou Jia[*],** Tuo An[*], Xinying Guo[*], S. Leng, H. Geng, Y. Ze, T. Harada, P. Torr, O. Mees, M. Pollefeys, Z. Liu, J. Wu, P. Abbeel, J. Malik, Y. Du, J. Yang

Learning-based observer for coupled disturbance — ICRA'26
Jindou Jia[*], Meng Wang[*], Z. Yang, B. Yang, Y. Liu, K. Guo, X. Yu

Agile flight control under multiple disturbances for quadrotor: Algorithms and evaluation — T-AES'22
Jindou Jia, K. Guo, X. Yu, L. Guo, L. Xie

Flight control for quadrotor safety in the presence of CoG shift and loss of motor efficiency — ICUAS'22 Oral Presentation
Jindou Jia, K. Guo, C. Wang, N. Shen, W. Jia, X. Yu

Multiple observers-based anti-disturbance control for a quadrotor UAV against payload and wind disturbance — CEP'20
Kexin Guo[*], **Jindou Jia[*],** X. Yu, L. Guo, L. Xie

Dual-disturbance observers-based control of UAV subject to internal and external disturbances — CAC'19 First Prize of Outstanding Paper
Kexin Guo[*], **Jindou Jia[*],** X. Yu, L. Guo

Preprints

ODE policy: Robot learning via action dynamics, not actions — arXiv'26
Jindou Jia[*], Jingliang Li[*], Kuangji Zuo[*], Tuo An[*], Y. Hu, M. Zhao, B. Lyu, J. Bai, D. Zhu, H. Dai, J. Zhang, J. Yang[†]

MARS policy: Multimodality only when it matters — arXiv'26
Jindou Jia[*], Tuo An[*], Yuxuan Hu[*], G. Li, J. Li, B. Hou, X. Chen, J. Bai, B. Lyu, J. Yang

MOSS: Mode-oriented source shaping for robot flow matching policies — arXiv'26
Zirun Ma[*], **Jindou Jia[*],** T. An, J. Bai, J. Dong, J. Yang

Δ-Flow matching for robot foundation models — arXiv'26
K. Zuo, T. An, X. Guo, G. Li, M. Zhao, J. Bai, B. Lyu, Z. Yu, J. Yang, J. Li, **Jindou Jia[†],** J. Yang[†]

TOPS: Reshaping flow source and transport for multi-task robot manipulation — arXiv'26
T. An, C. Zhou, J. Dong, J. Li, J. Bai, **Jindou Jia[†],** J. Yang[†]

TUCO: Curating simulation demonstrations for sim-to-real robot policy co-training arXiv'26
N. Zhu, M. Zhao, Y. Tang, Z. Sun, P. Li, D. Ni, **Jindou Jia**[†], J. Yang[†]

APEX: Adaptive policy execution for precise manipulation arXiv'26
M. Zhao, C. Jiang, T. An, **Jindou Jia**[†], J. Yang[†]

Optimizing control-friendly trajectories with self-supervised residual learning arXiv'26
K. Guo, Z. Yang, Y. Liu, **Jindou Jia**[†], X. Yu

Main Awards and Honors

2024 ICCA Best Student Paper Award — 2024, IEEE International Conference on Control and Automation
Title: *TRACE - Trajectory refinement with control error enables safe and accurate maneuvers.*

2019 CAC First Prize of Outstanding Paper — 2019, IEEE Chinese Automation Conference
Title: *Dual-disturbance observers-based control of UAV subject to internal and external disturbances.*

Outstanding graduate of Beihang University — 2025, Beihang, China

Academic Excellence Fund for Doctoral Candidate — 2024 - 2025, Beihang, China
Funds, ⩽1%, Project leader

Outstanding Research Project of Shenyuan Honors College — 2022 - 2024, Beihang, China
Funds, Project leader

Four-times Outstanding Academic Innovation Award — 2022 - 2023, Beihang, China

Two-times First-class Scholarship — 2021 - 2022, Beihang, China

China Classification Society Scholarship — 2017, China Classification Society

First prize in Mathematical Contest in Modeling — 2016, Heilongjiang Province

First prize in the Chinese Mathematics Competitions — 2016, China

National Scholarship — 2016, China

Talks and Media

Classical Control and Planning Meet Modern Learning — Nov 2026, Guest Lecture
CEG5304 Deep Learning for Digitalization Technologies · National University of Singapore (NUS)

Classical Control and Planning Meet Modern Learning — Oct 2026, Invited Talk
Hosted by Prof. Shangke Lyu · Nanjing University (NJU)

Robopapers Podcast — Aug 2026, Interview
Guest interview on A2A policy

Seminar on Embodied AI Robotics — Jun 2026, Invited Talk
Efficient robot policy learning · SIMTech & ARTC, Singapore

TechBeat — Mar 2025, Invited Talk
Invited technical talk on feedback neural ODE

Professional Service

Associate Editor, IEEE Robotics and Automation Letters (RA-L)

Reviewer, T-RO, IJRR, RA-L, ICRA, and other robotics venues